

SERS Substrate-Agnostic Dataset Matrix

The current substrate-agnostic evaluation uses canonical chemical labels. In this analysis, BT is merged into benzenethiol. The table below shows the current number of spectra for each chemical-substrate pair.

Table 1: Current canonical three-chemical SERS matrix.

Chemical	Ag	AgNP	Au	AuNP	PICO	pSERS	Current total
4np	0	25	0	0	25	25	75
benzenethiol	25	0	25	0	25	25	100
pyridine	0	25	0	25	25	25	100

The minimum recommended expansion is to fill every missing chemical-substrate cell before heavily increasing spectra from already-covered cells.

Table 2: Minimum target matrix for stronger substrate-agnostic evaluation.

Chemical	Ag	AgNP	Au	AuNP	PICO	pSERS	Target total
4np	25+	25+	25+	25+	25+	25+	150+
benzenethiol	25+	25+	25+	25+	25+	25+	150+
pyridine	25+	25+	25+	25+	25+	25+	150+

If time allows, the stronger target is two to three independent preparations/maps per chemical-substrate pair, with approximately 25 spectra per preparation. Independent preparations are more useful for substrate-agnostic validation than many correlated spectra from a single map.